

Proposed Solution

HDI Prediction System — Project Design Phase

Overview

The proposed solution is a web-based HDI Prediction System that estimates a country's Human Development Index using a trained regression model. Users input four core development indicators and receive an instant predicted HDI score through a clean, dark-themed web interface.

Proposed Approach

- Collect and clean a 2021 global HDI dataset covering approximately 195 countries.
- Engineer four core features: Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita.
- Train a Linear Regression model on these features against the actual HDI target, achieving an R^2 score of approximately 0.98.
- Serialize the trained model and integrate it into a Flask backend.
- Build a simple, responsive front-end form for indicator input and result display.
- Deploy the complete application on Render.com for public access.

Unique Selling Points (USP)

- Instant, on-demand HDI prediction — no manual calculation or waiting for annual reports.
- Lightweight and interpretable model, keeping the solution fast and easy to maintain.
- Simple four-field input, making it accessible to non-technical users.
- Fully deployed and publicly accessible web application, not just a notebook demo.

Value Proposition

By converting a well-known but computation-heavy development metric into a single-click prediction tool, the solution saves time for students, researchers, and analysts, and makes development-economics concepts more approachable and explorable for a broader audience.

Scalability

- The four-feature model can be extended with additional years of HDI data as they become available.
- The same Flask architecture can support additional prediction endpoints (e.g. regional HDI trends) with minimal changes.
- Cloud deployment on Render.com allows the application to scale with user demand without infrastructure management overhead.